MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

RELAY, LATCHING, KEY CODE MFD

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic latching relay module.

1.2 Classification. This specification covers one class of modules.

Class I as specified in MIL-STD-1389.

2. APPLICABLE DOCUMENTS

2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.2 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Naval Electronic Systems Command, ATTN: ELEX 8111, Department of the Navy, Washington, DC 20363, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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3.3 Failure rate requirements. All modules acquired to this specification shall have a predicted failure rate no greater than 0.433 failures per million operating hours when predicted in accordance with MIL-STD-1389.

3.4 Performance requirements. The module shall conform to the electrical requirements specified in table I and shall operate within the environment of design class I as specified in MIL-STD-1389.

3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.5.1 Key code. The module key code shall be MFD.

3.5.2 Size. The module size shall be 1B.

3.5.3 Weight. The module shall have a maximum weight of 70 grams.

3.5.4 Contact pins. The module shall have 40 contact pins positioned A1 through A20 and B21 through B40.

3.5.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.5.6 Depth. The module shall have no additional depth requirements.

3.5.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.5.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/343-1, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier).

3.6 Circuit pin configuration. The function diagram and pin assignments shall conform to figure 1.

3.7 Acceptance tests. All production modules shall be tested in accordance with footnotes 1 and 2 of table I.

3.8 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table I (see 6.2).

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

4.3 Test equipment. Test equipment specified on figures and in tables with a type designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified.

4.4 Test procedures. The module shall be tested as specified in table I.

4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Program Quality Assurance Activity and they conform to the requirements of MIL-M-28787.

4.4.2 Order of testing. Test conditions A through K shall be performed in any order with the following exceptions. If test conditions A or B are part of a sequence of tests, the order of testing shall be as follows: A, B where applicable, followed by the remaining tests in the sequence. If test conditions A or B are performed during quality assurance inspection testing, they shall be followed by performance of all the tests in table I which are marked with an "X" in the test type column.

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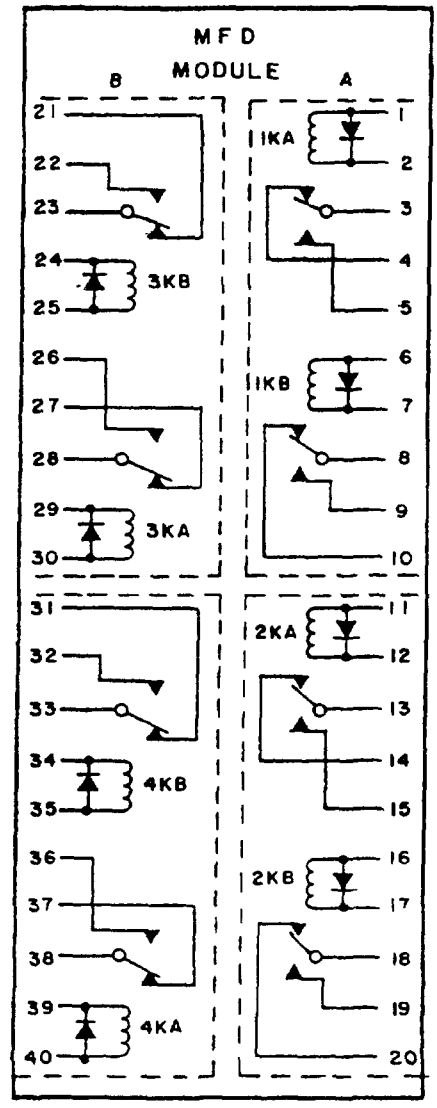


FIGURE 1. MFD function diagram, latching relay.

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TABLE I. Module electrical requirements.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25°C ±5°C ambient temperature				Operating temperature 0°C to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
A1-A15	Isolation resistance	II	X	10	∞			10	∞			MΩ
A16-A20	Insulation resistance	II	X	10	∞			10	∞			MΩ
B1-B16	Resistance, open contact	III	X	10	∞			10	∞			MΩ
C1-C16	Resistance, closed contact	IV	X	0.00	0.13	0.00	0.20	0.00	0.13	0.00	0.20	Ω
D1-D8	Resistance, relay coil	V	X	1500	2500	1500	2500	1400	2600	1400	2600	Ω
E1-E8	Forward voltage drop, diode	VI	X	0	800	0	800	0	800	0	800	mV
F1-F16	Pick-up voltage, relay	VII	X	9	19	9	19	9	19	9	19	V
G1-G16	Operate time	VIII	S	0.0	1.5	0.0	2.5	0.0	1.5	0.0	2.5	ms
H1-H16	Contact bounce	IX	S	0	2	0	3	0	2	0	3	ms
J1-J16	Contact carry current check	X	Q	0	50	0	100	0	50	0	100	mV
K1-K16	Intercontact capacitance	XI	Q	0	5	0	5	0	5	0	5	pF

1/ "X" parameters shall be 100 percent tested during module acceptance tests.

2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.

3/ "Q" parameters shall be tested during qualification testing only.

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4.5 Qualification testing. Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through K in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25°C ±5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions A through K except A20 shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 2.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 3.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be nonoperating. Following the vibration tests, the electrical tests of table I which are marked with an "X" in the test type column shall be performed using the initial limits at 25°C ±5°C ambient.

4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 3. Following the life test and while still at the maximum class temperature, electrical tests A through K except A20 shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements of table I. Following the return of the module to 25°C, electrical tests A through K shall be performed in the order specified in 4.4.2 using the 25°C ±5°C ambient end-of-life requirements of table I.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

6. NOTES

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

6.2 End-of-life When any individual module circuit parameter fails to conform to the end-of-life requirements of table I, the module is considered to have completed its useful life (see 3.8).

6.3 Module description. The following electrical parameters reflect the 0°C to 60°C end-of-life limits from table I:

Maximum power dissipation	1.8 W
Relay coil voltage	32 V maximum
Relay coil voltage	26.5 V nominal
Relay coil resistance	2000Ω ±30%
Operating voltage	19 V minimum
Minimum operate time	2.5 ms at nominal rated coil voltage
Minimum operate pulse	3.5 ms at nominal rated coil voltage
Contact bounce	3 ms maximum

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Contact current load rating	DC - 1 A at 28 V dc resistive load AC - 600 mA at 115 V, 400 Hz, 400 mA at 115 V, 60 Hz.
Contact carry current	5 A (continuous, unswitched)
Contact resistance	Initial 0.1Ω maximum; 0.2Ω after life
Intercontact capacitance	5 pF maximum
Contact life rating	10^7 operations typical at low level; 10^6 operations, minimum at 0.5 A, 28 V dc; 10^5 operations, minimum at all other loads specified above

WARNING

Relay coils should not be energized simultaneously with either ac or dc voltages. Particular attention should be given to transients as a short pulse above rated voltage applied to both coils or to one coil with the other energized may cause permanent damage.

6.4 Module function. This module contains four independent double pole double throw magnetic latching relays with memory capability. Relays may be operated by a short duration pulse and will not reset upon removal of the power. Set and reset is obtained by alternate operation of coils KA and KB.

Agent:
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Preparing activity:
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(Project 5963-N433)

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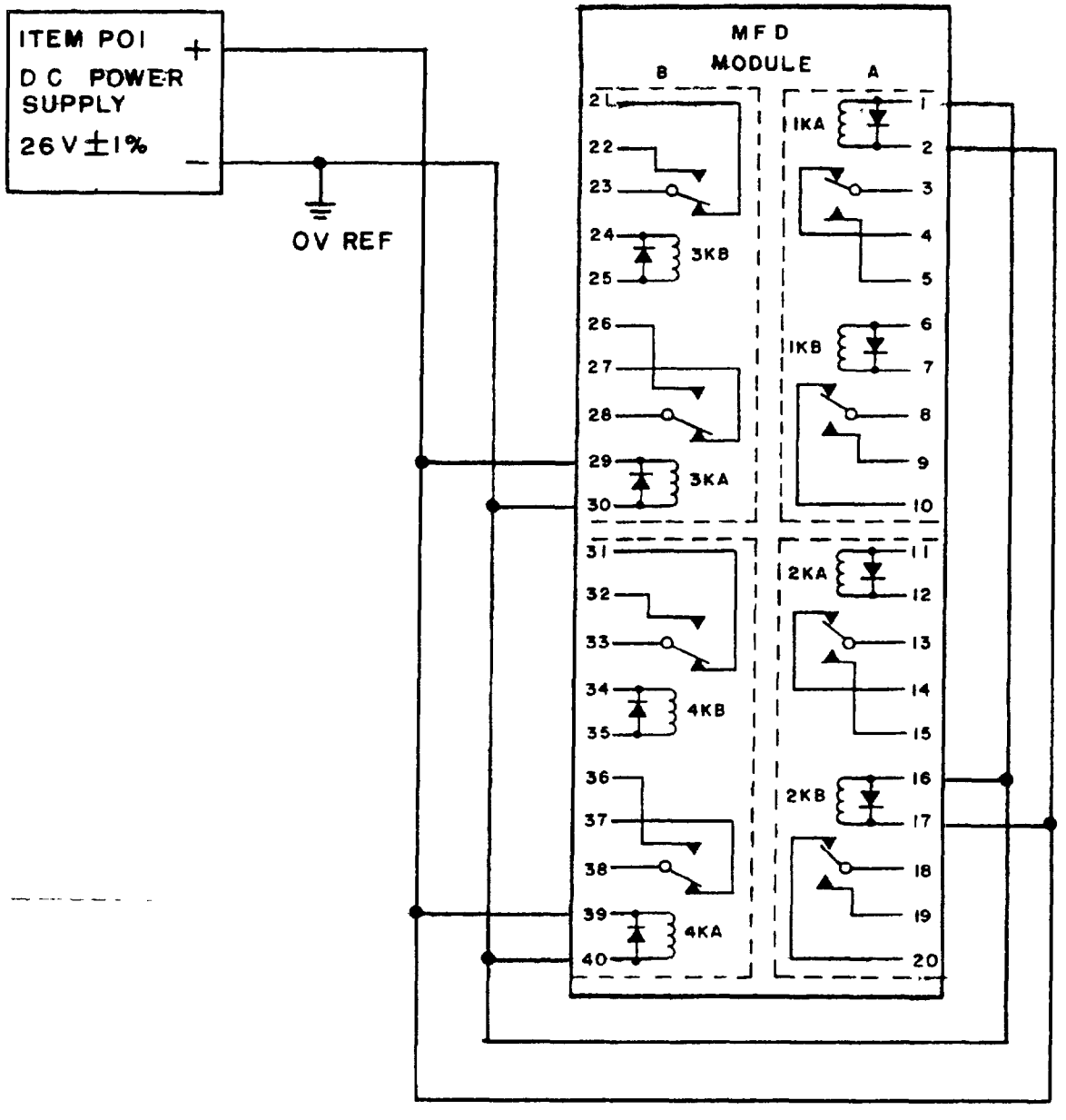


FIGURE 2. Operating temperature and vibration test circuit.

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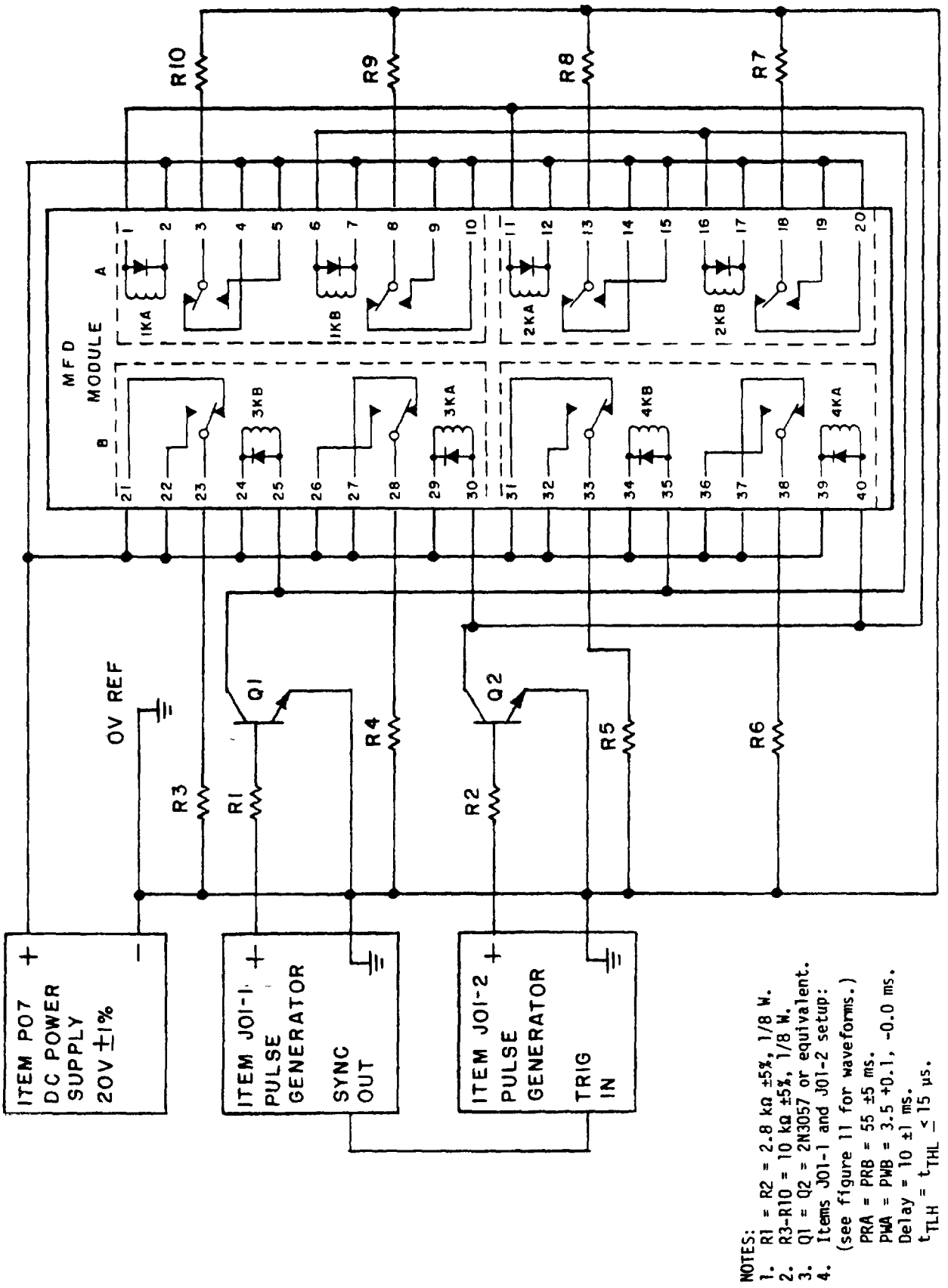


FIGURE 3. Component temperature and life test circuit.

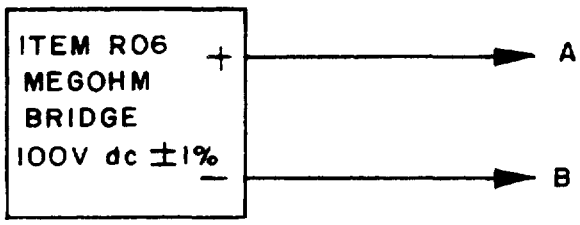
NOTES:
1. R1 = R2 = 2.8 kΩ ±5%, 1/8 W.
2. R3-R10 = 10 kΩ ±5%, 1/8 W.
3. Q1 = Q2 = 2N3057 or equivalent.
4. Items J01-1 and J01-2 setup:
   (see figure 11 for waveforms.)
   PRA = PRB = 55 ±5 ms.
   PWA = PWB = 3.5 +0.1, -0.0 ms.
   Delay = 10 ±1 ms.
   $t_{TLH} = t_{THL} \leq 15$ µs.

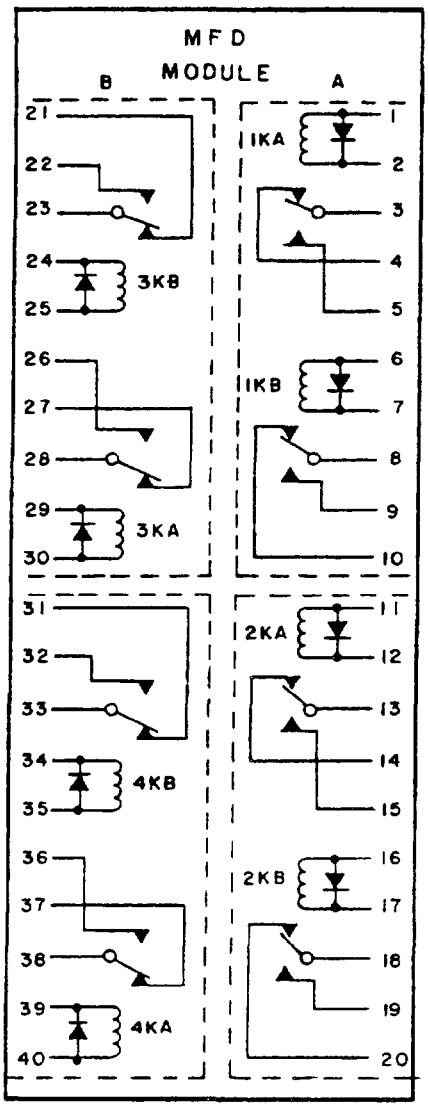
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TABLE II. Isolation and insulation resistance test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
A1	Isolation resistance	1-20	21-40	10	∞	MΩ	4
A2	"	1-10	11-20	"	"	"	"
A3	"	21-30	31-40	"	"	"	"
A4	"	21-23	26-28	"	"	"	"
A5	"	31-33	36-38	"	"	"	"
A6	"	3-5	8-10	"	"	"	"
A7	"	13-15	18-20	"	"	"	"
A8	Isolation resistance, coil contacts	1,2	3-5,8-10	"	"	"	"
A9	"	6,7	3-5,8-10	"	"	"	"
A10	"	11,12	13-15,18-20	"	"	"	"
A11	"	16,17	13-15,18-20	"	"	"	"
A12	"	24,25	21-23,26-28	"	"	"	"
A13	"	29,30	21-23,26-28	"	"	"	"
A14	"	34,35	31-33,36-38	"	"	"	"
A15	"	39,40	31-33,36-38	"	"	"	"
A16	Insulation resistance, coil-coil	1,2	6,7	"	"	"	"
A17	"	11,12	16,17	"	"	"	"
A18	"	24,25	29,30	"	"	"	"
A19	"	34,35	39,40	"	"	"	"
A20	Insulation resistance, frame	1-20,21-40	Frame	"	"	"	"

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WARNING

When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "discharge" before connecting or disconnecting the test leads.

NOTE. For test condition A20, penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal.

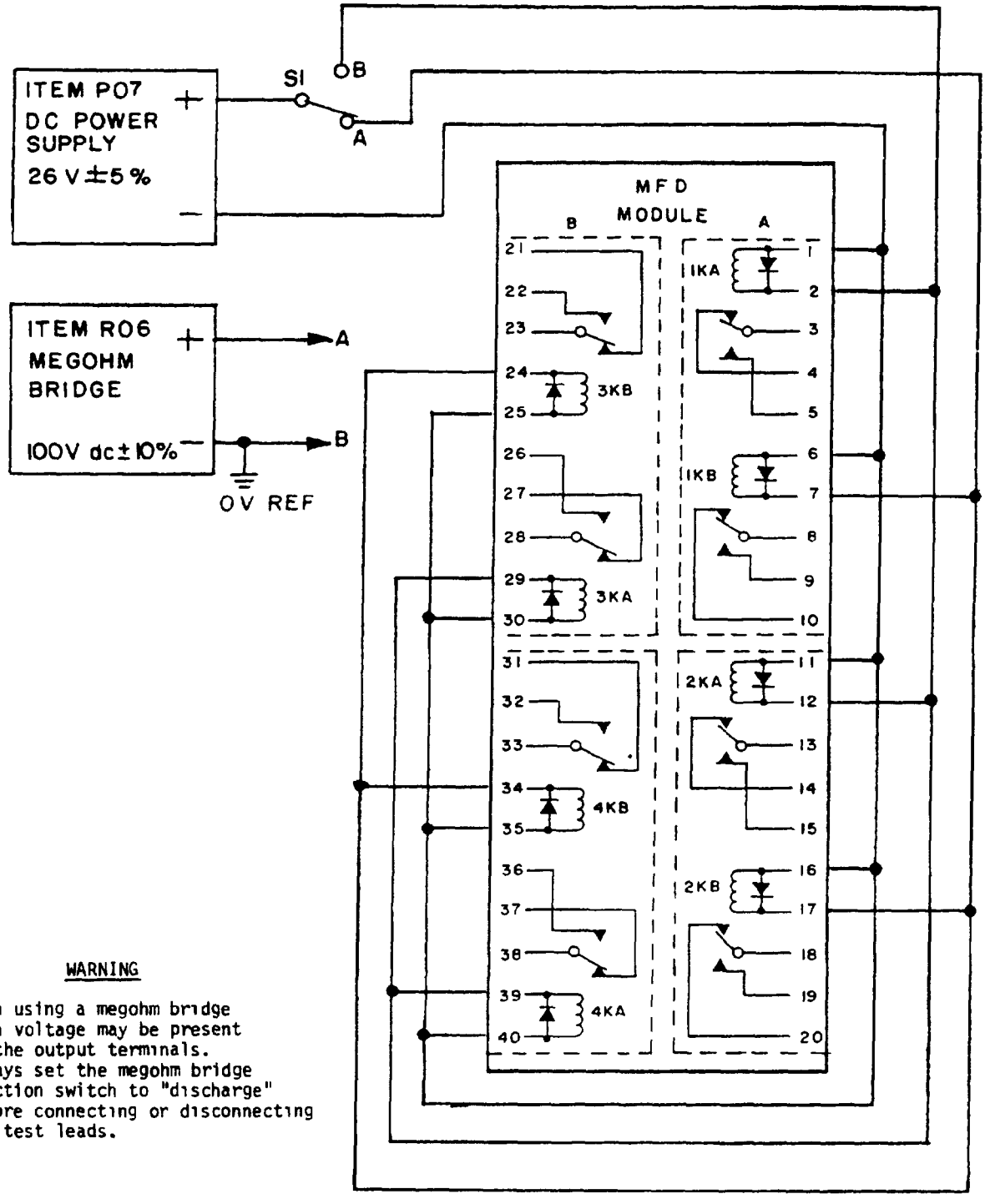
FIGURE 4. Isolation and insulation resistance test circuit for table II.

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TABLE III. Resistance (open contact) test.

Test condition	Parameter		Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
B1	1KA	Resistance, open contact	3	5	10	∞	MΩ	5
B2	1KB	"	8	9	"	"	"	"
B3	2KA	"	13	15	"	"	"	"
B4	2KB	"	18	19	"	"	"	"
B5	3KB	"	23	22	"	"	"	"
B6	3KA	"	26	28	"	"	"	"
B7	4KB	"	33	32	"	"	"	"
B8	4KA	"	36	38	"	"	"	"
B9	1KA	"	3	4	"	"	"	"
B10	1KB	"	8	10	"	"	"	"
B11	2KA	"	13	14	"	"	"	"
B12	2KB	"	18	20	"	"	"	"
B13	3KB	"	21	23	"	"	"	"
B14	3KA	"	27	28	"	"	"	"
B15	4KB	"	31	33	"	"	"	"
B16	4KA	"	37	38	"	"	"	"

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WARNING

When using a megohm bridge high voltage may be present at the output terminals. Always set the megohm bridge function switch to "discharge" before connecting or disconnecting the test leads.

NOTES
1. S1 = SPDT switch.
2. Place switch S1 in position A for test conditions B1 through B8 and in position B for test conditions B9 through B16.

FIGURE 5. Resistance (open contact) test circuit for table III.

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TABLE IV. Resistance (closed contact) check test.

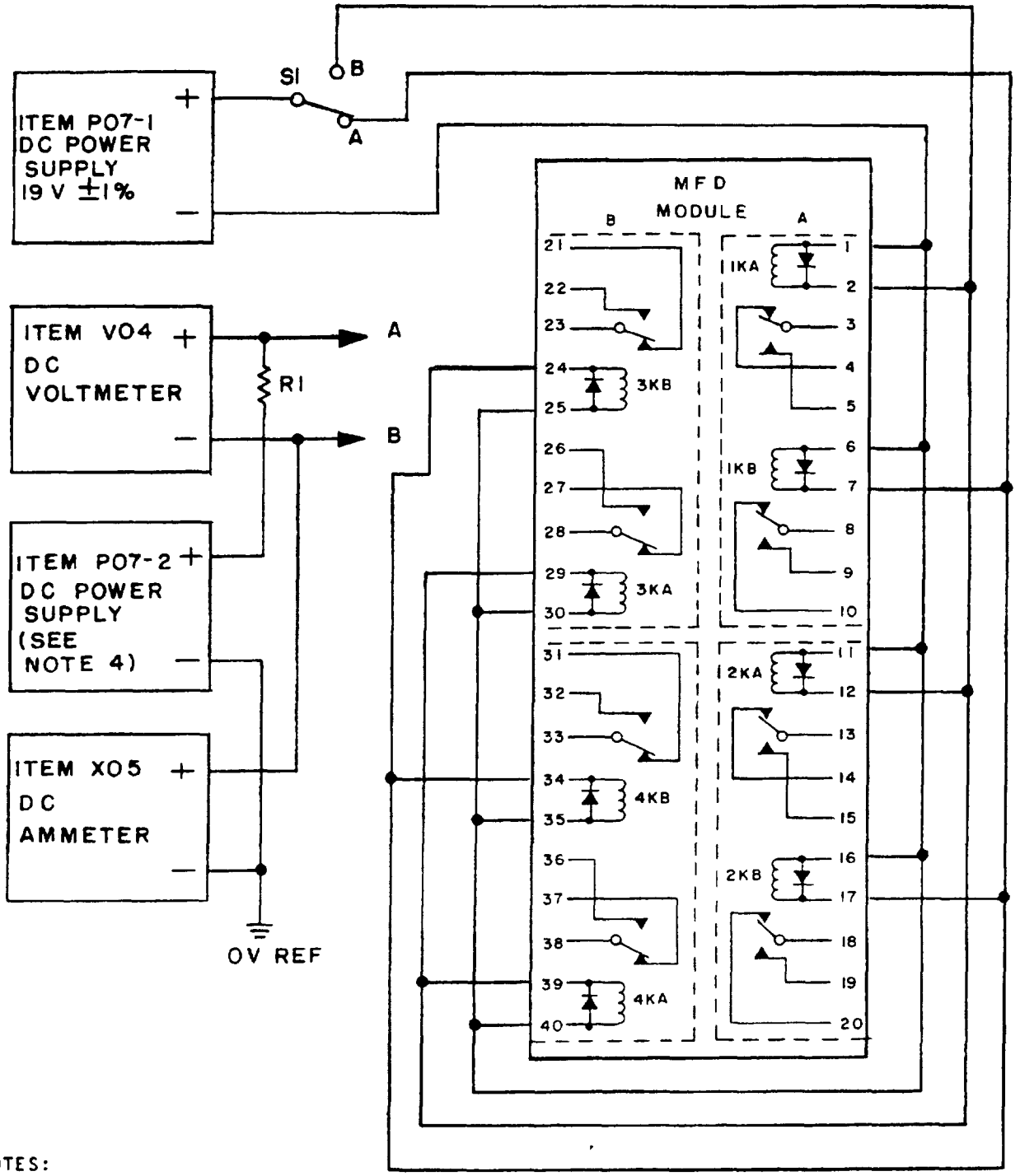
Test condition	Parameter		Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
C1	1KA	Resistance, closed contact, check	3	4	0	13	mV	6
C2	1KB	"	8	10	"	"	"	"
C3	2KA	"	13	14	"	"	"	"
C4	2KB	"	18	20	"	"	"	"
C5	3KB	"	21	23	"	"	"	"
C6	3KA	"	27	28	"	"	"	"
C7	4KB	"	31	33	"	"	"	"
C8	4KA	"	37	38	"	"	"	"
C9	1KA	"	3	5	"	"	"	"
C10	1KB	"	8	9	"	"	"	"
C11	2KA	"	13	15	"	"	"	"
C12	2KB	"	18	19	"	"	"	"
C13	3KB	"	22	23	"	"	"	"
C14	3KA	"	26	28	"	"	"	"
C15	4KB	"	32	33	"	"	"	"
C16	4KA	"	36	38	"	"	"	"

1/ The closed contact resistance in ohms can be determined from the equation:

$$R = \frac{V}{I} = \frac{V}{100}.$$

V = the measured voltage across the relay contacts in mV.
I = the current through the contact = 100 mA.

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NOTES:
1. R1 = 50Ω ±1%, 1 W metal film.
2. S1 = SPDT switch.
3. Place switch S1 in position A for test conditions C1-C8 and in position B for test conditions C9-C16.
4. Adjust item P07-2 until item X05 reads 100 ±1 mA (approximately 5 V).
5. Voltage sensing leads from item V04 shall be connected directly to the module pins (within 0.5 inch (12.7 mm) maximum) to exclude the resistance of the current carrying leads from item P07-2.

FIGURE 6. Resistance (closed contact) check test circuit for table IV.

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TABLE V. Resistance (relay coil) test.

Test condition	Parameter		Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
D1	1KA	Resistance, relay coil	2	1	1500	2500	Ω	7
D2	1KB	"	7	6	"	"	"	"
D3	2KA	"	12	11	"	"	"	"
D4	2KB	"	17	16	"	"	"	"
D5	3KB	"	24	25	"	"	"	"
D6	3KA	"	29	30	"	"	"	"
D7	4KB	"	34	35	"	"	"	"
D8	4KA	"	39	40	"	"	"	"

TABLE VI. Forward voltage drop (diode) test.

Test condition	Parameter		Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
E1	1KA	Forward voltage drop, diode	1	2	0	800	mV	8
E2	1KB	"	6	7	"	"	"	"
E3	2KA	"	11	12	"	"	"	"
E4	2KB	"	16	17	"	"	"	"
E5	3KB	"	25	24	"	"	"	"
E6	3KA	"	30	29	"	"	"	"
E7	4KB	"	35	34	"	"	"	"
E8	4KA	"	40	39	"	"	"	"

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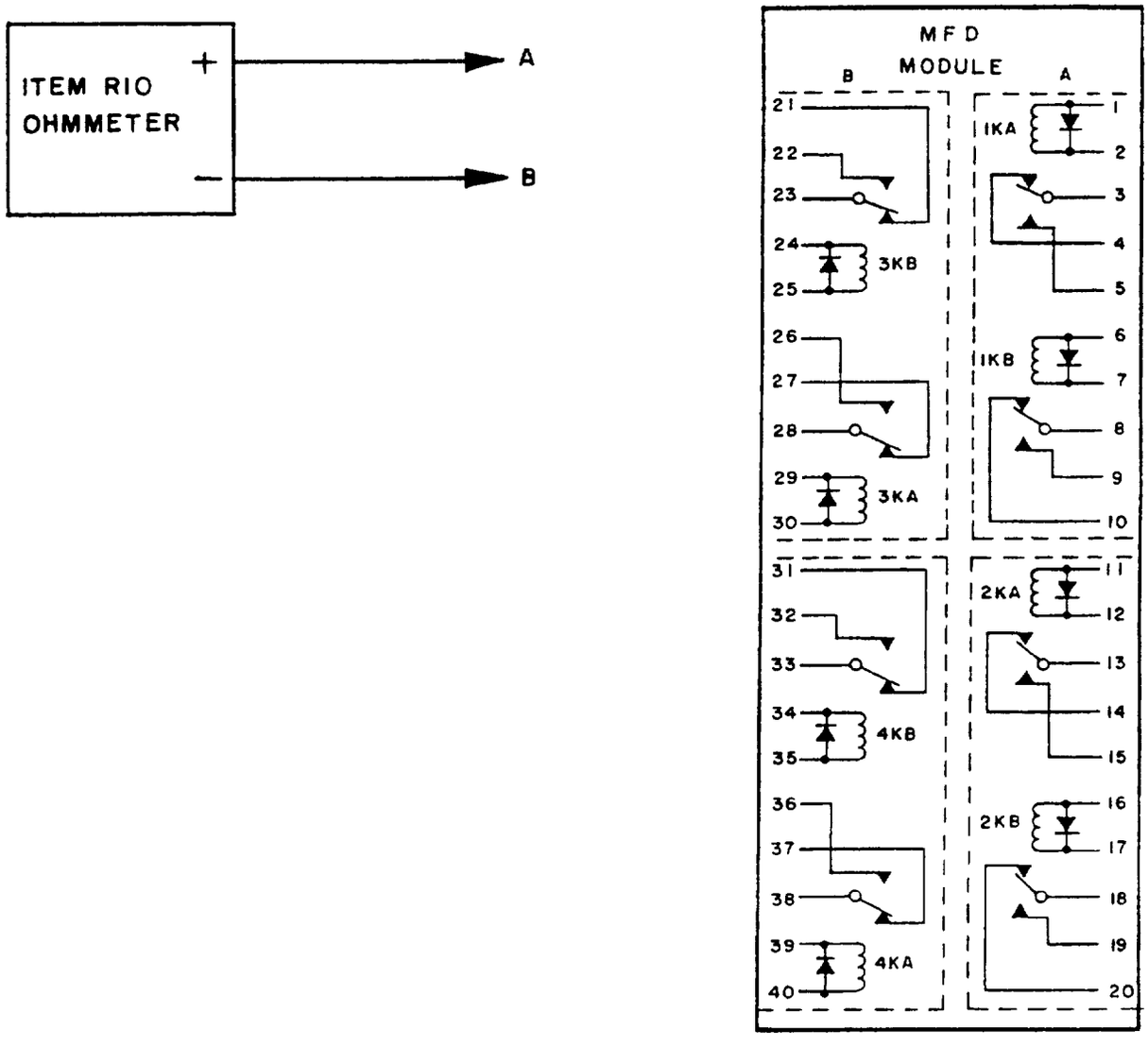
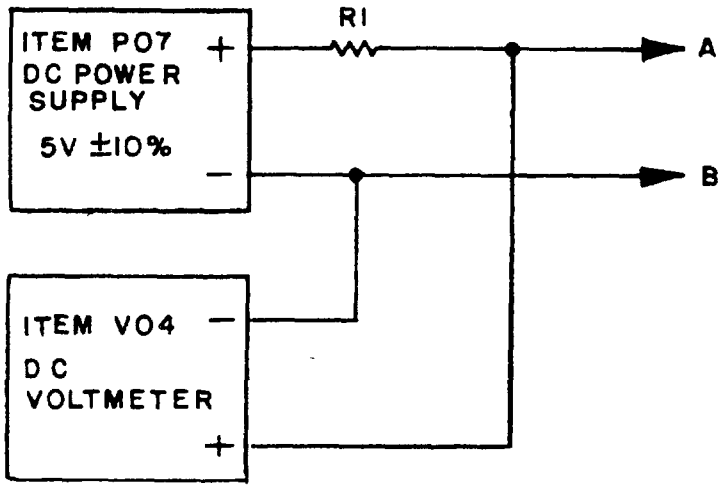
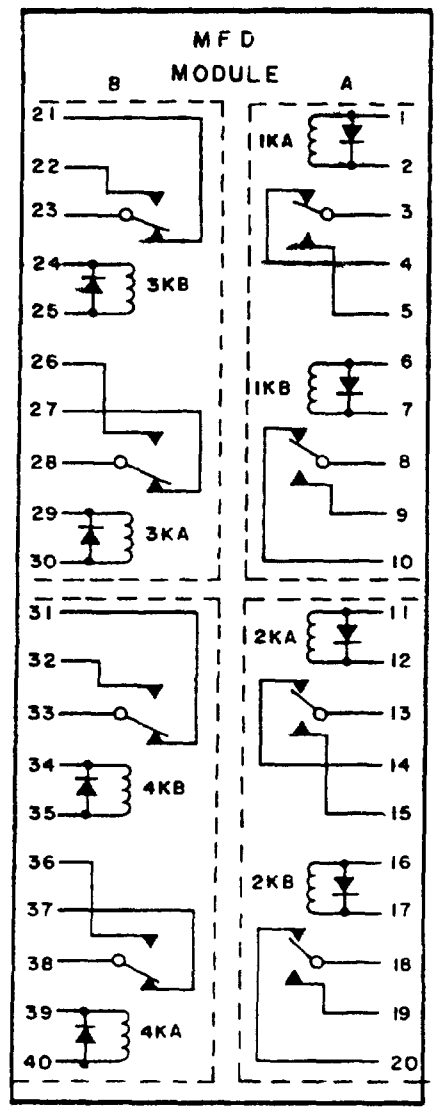


FIGURE 7. Resistance (relay coil) test circuit for table V.

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NOTE: R1 = 430Ω ±1%, 1/8 W metal film.

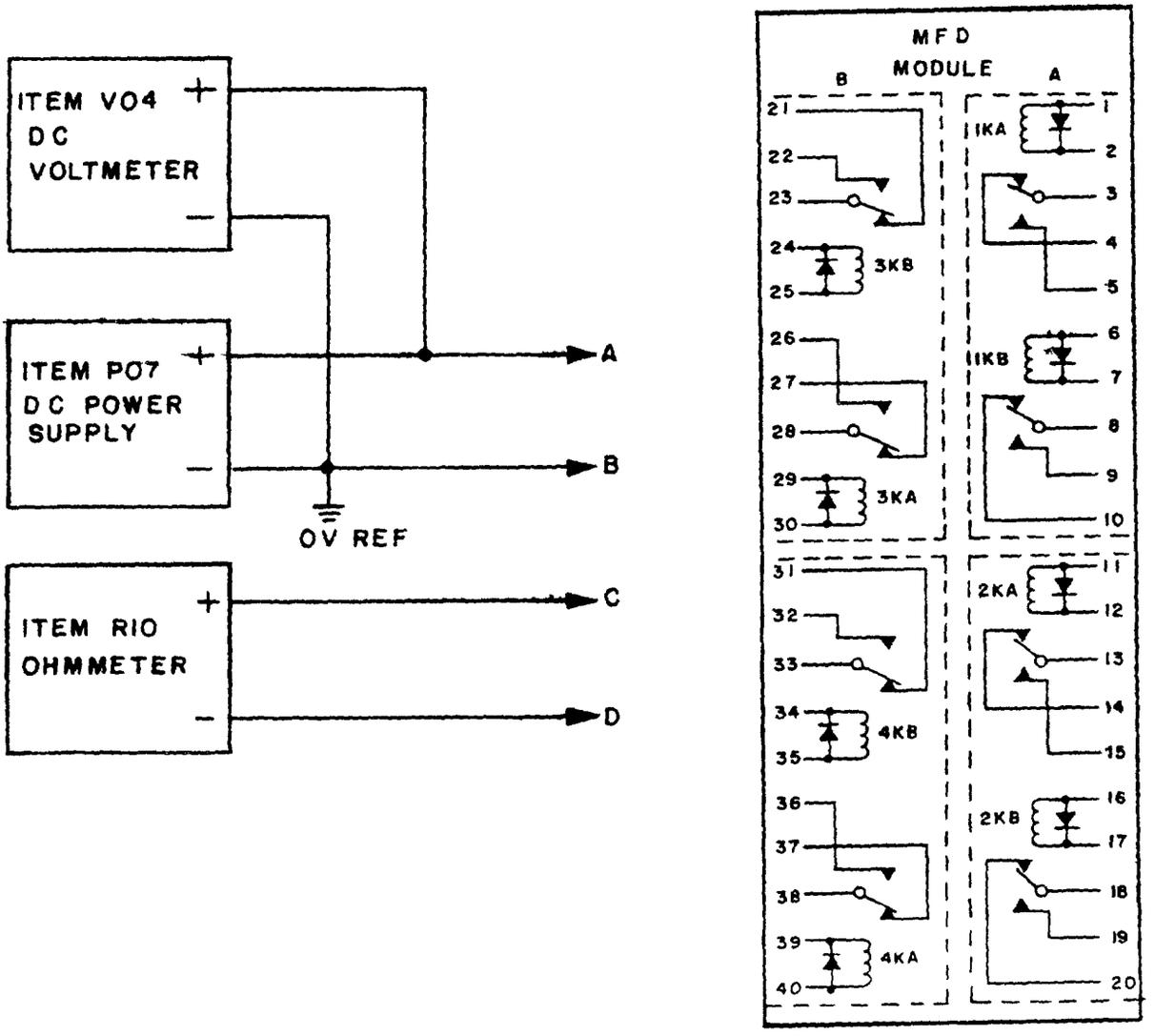
FIGURE 8. Forward voltage drop (diode) test circuit for table VI.

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TABLE VII. Pick-up voltage (relay) test.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
---	Initial set-up condition	7,17,24, 34	6,16,25, 35	Open	Open	---	---	---	9
F1	1KA Pick-up voltage, relay	2	1	3	5	9	19	V	"
F2	1KB "	7	6	3	4	"	"	"	"
F3	1KA "	2	1	8	9	"	"	"	"
F4	1KB "	7	6	8	10	"	"	"	"
F5	2KA "	12	11	13	15	"	"	"	"
F6	2KB "	17	16	13	14	"	"	"	"
F7	2KA "	12	11	18	19	"	"	"	"
F8	2KB "	17	16	18	20	"	"	"	"
F9	3KA "	29	30	28	26	"	"	"	"
F10	3KB "	24	25	28	27	"	"	"	"
F11	3KA "	29	30	23	22	"	"	"	"
F12	3KB "	24	25	23	21	"	"	"	"
F13	4KA "	39	40	38	36	"	"	"	"
F14	4KB "	34	35	38	37	"	"	"	"
F15	4KA "	39	40	33	32	"	"	"	"
F16	4KB "	34	35	33	31	"	"	"	"

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NOTES:
1. See table VII for initial setup. Set item P07 to 19 V.
2. Increase item P07 voltage slowly from zero until item R10 indicates a closed contact (< 1Ω). Note voltage of item P07 on closure of contacts as indicated on item R10.

FIGURE 9. Pick-up voltage (relay) test circuit for table VII.

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TABLE VIII. Operate time test.

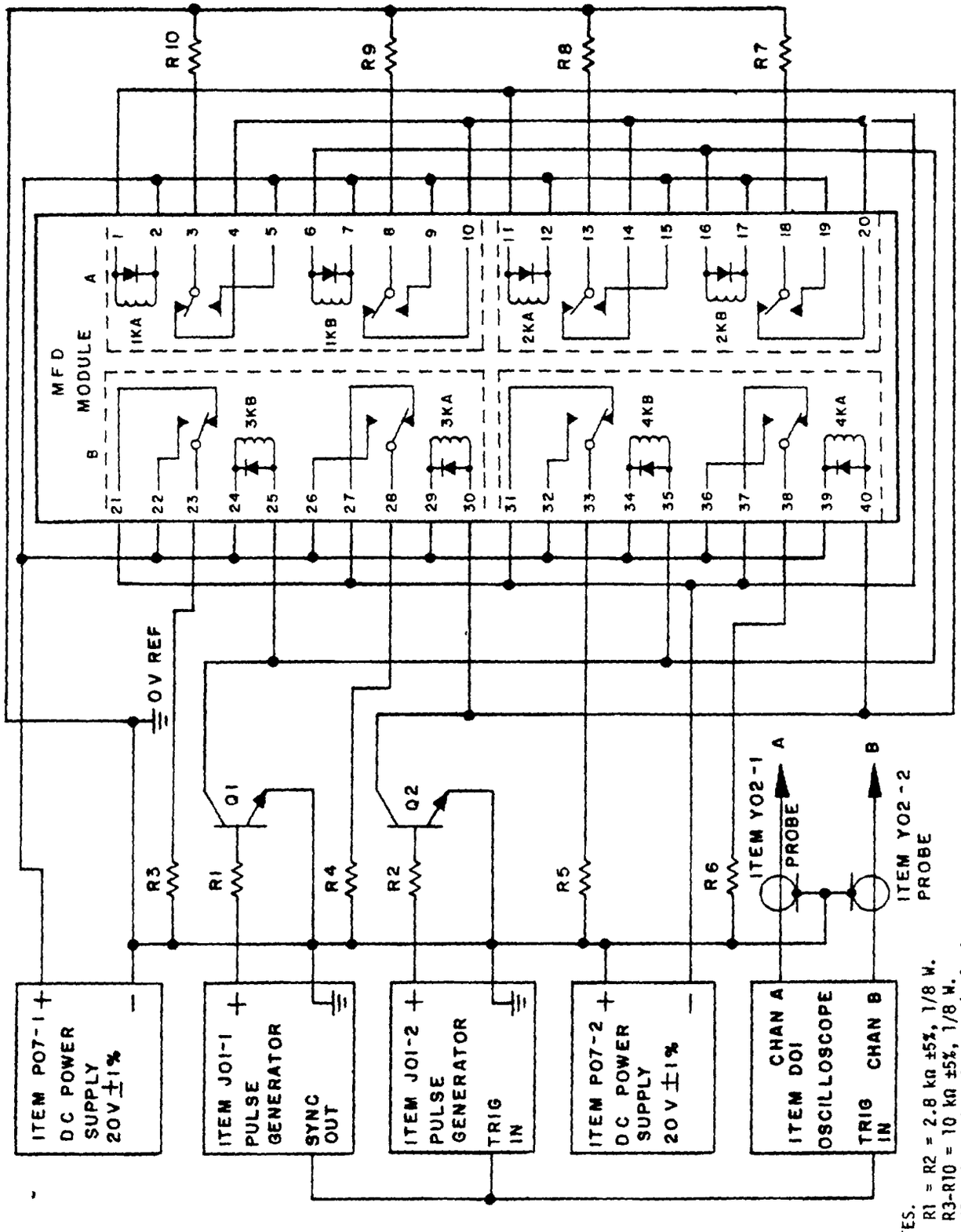
Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to	B to pin	Min	Max		
G1	1KA Operate time (t_o)	Item J01-1	3	0.0	1.5	ms	10,11
G2	1KB "	"	8	"	"	"	"
G3	2KA "	"	13	"	"	"	"
G4	2KB "	"	18	"	"	"	"
G5	3KB "	"	23	"	"	"	"
G6	3KA "	"	28	"	"	"	"
G7	4KB "	"	33	"	"	"	"
G8	4KA "	"	38	"	"	"	"
G9	1KA "	Item J01-2	3	"	"	"	"
G10	1KB "	"	8	"	"	"	"
G11	2KA "	"	13	"	"	"	"
G12	2KB "	"	18	"	"	"	"
G13	3KB "	"	23	"	"	"	"
G14	3KA "	"	28	"	"	"	"
G15	4KB "	"	33	"	"	"	"
G16	4KA "	"	38	"	"	"	"

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TABLE IX. Contact bounce test.

Test condition	Parameter		Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to	B to pin	Min	Max		
H1	1KA	Contact bounce (t_b)	Item J01-1	3	0	2	ms	10,11
H2	1KB	"	"	8	"	"	"	"
H3	2KA	"	"	13	"	"	"	"
H4	2KB	"	"	18	"	"	"	"
H5	3KB	"	"	23	"	"	"	"
H6	3KA	"	"	28	"	"	"	"
H7	4KB	"	"	33	"	"	"	"
H8	4KA	"	"	38	"	"	"	"
H9	1KA	"	Item J01-2	3	"	"	"	"
H10	1KB	"	"	8	"	"	"	"
H11	2KA	"	"	13	"	"	"	"
H12	2KB	"	"	18	"	"	"	"
H13	3KB	"	"	23	"	"	"	"
H14	3KA	"	"	28	"	"	"	"
H15	4KB	"	"	33	"	"	"	"
H16	4KA	"	"	38	"	"	"	"

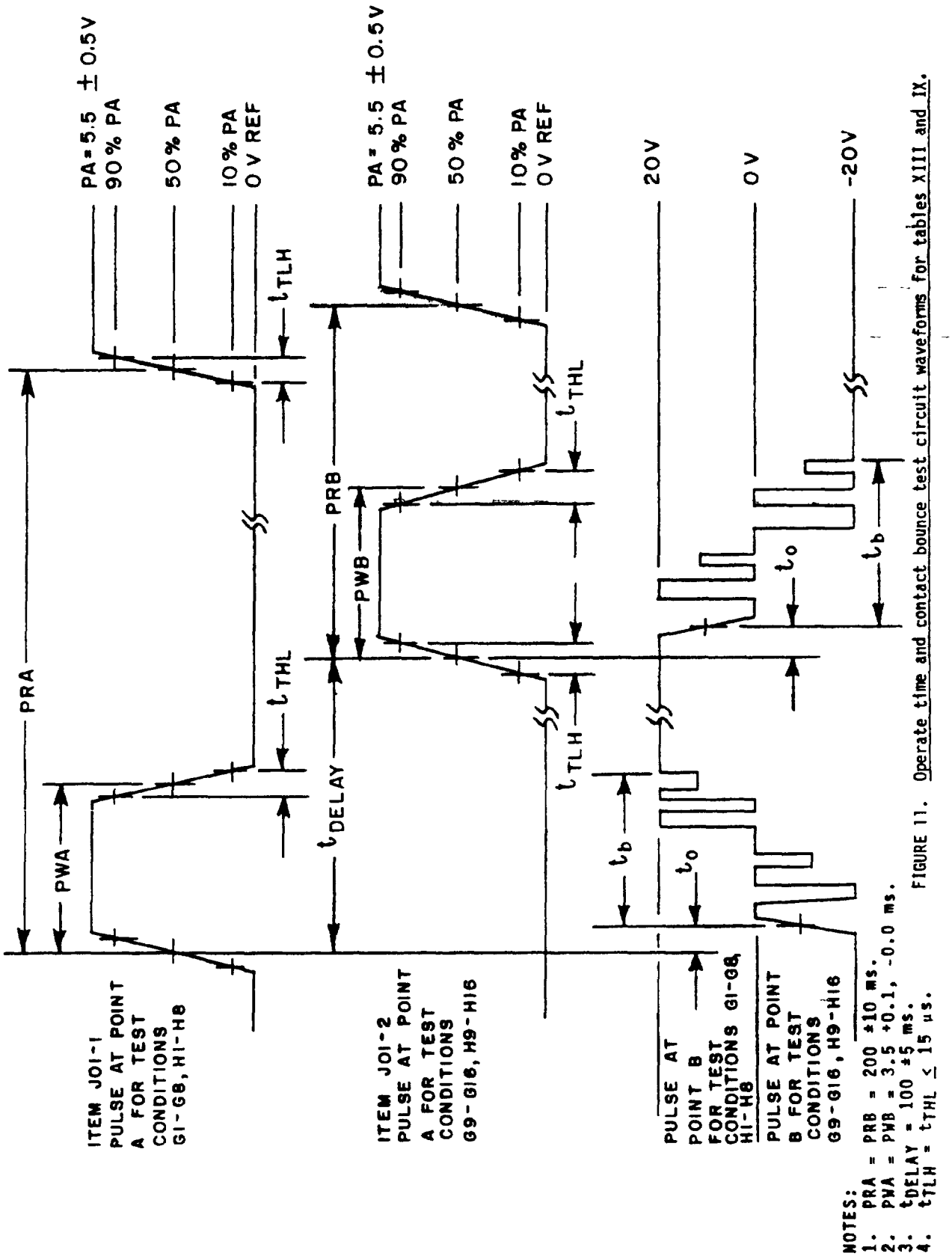
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NOTES.
1. R1 = R2 = 2.8 kΩ ±5%, 1/8 W.
2. R3-R10 = 10 kΩ ±5%, 1/8 W.
3. Q1 = Q2 = 2N3057 or equivalent.

FIGURE 10. Operate time and contact bounce test circuit for tables VIII and IX.

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NOTES:
1. PRA = PRB = 200 ±10 ms.
2. PWA = PWB = 3.5 +0.1, -0.0 ms.
3. t_{DELAY} = 100 ±5 ms.
4. $t_{TLH} = t_{THL} \leq 15$ µs.

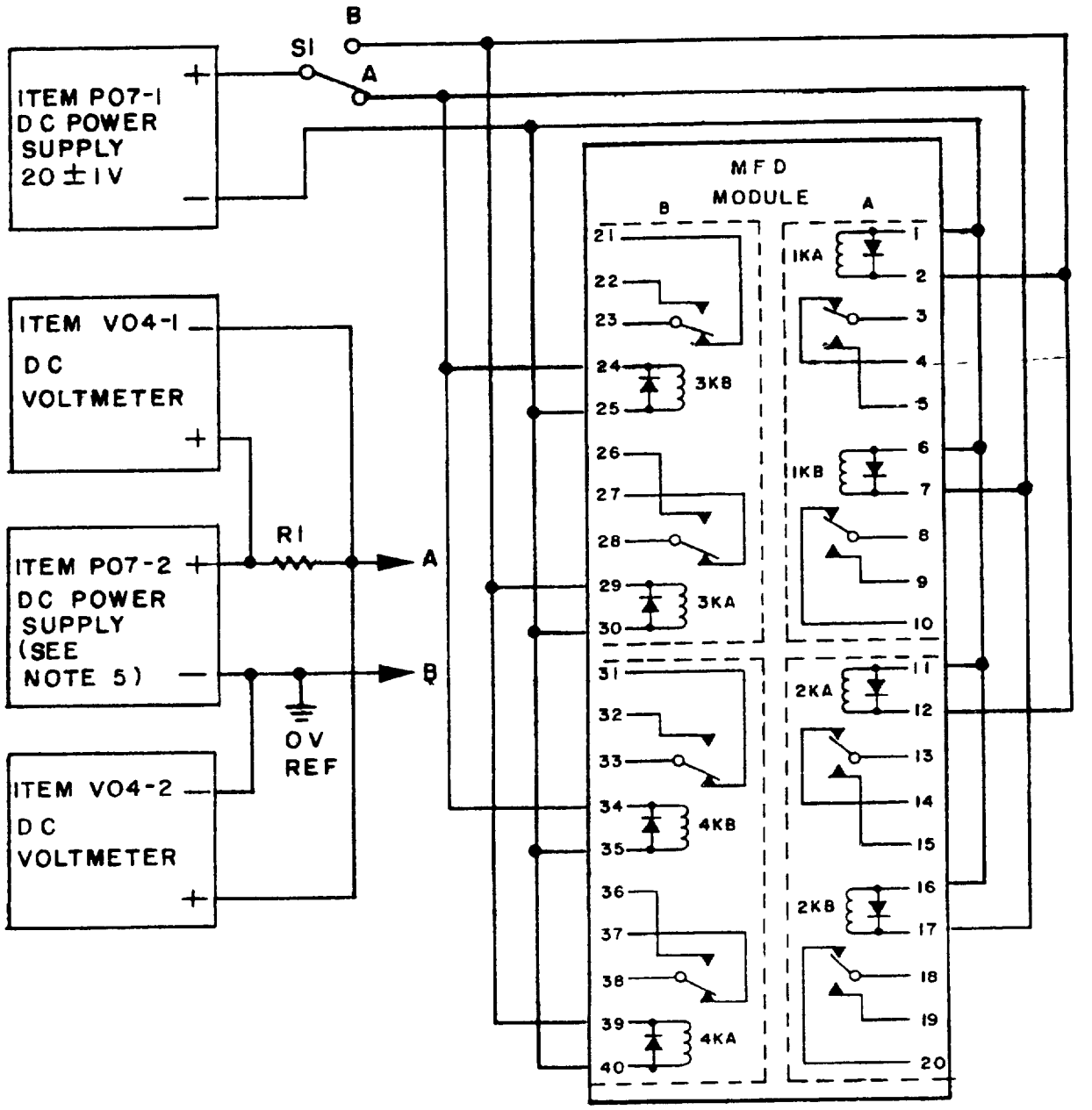
FIGURE 11. Operate time and contact bounce test circuit waveforms for tables XIII and IX.

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TABLE X. Contact carry current check test.

Test condition	Parameter		Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
J1	1KA	Contact carry current check	3	4	0	50	mV	12
J2	1KB	"	8	10	"	"	"	"
J3	2KA	"	13	14	"	"	"	"
J4	2KB	"	18	20	"	"	"	"
J5	4KA	"	38	37	"	"	"	"
J6	4KB	"	33	31	"	"	"	"
J7	3KA	"	28	27	"	"	"	"
J8	3KB	"	23	21	"	"	"	"
J9	1KA	"	3	5	"	"	"	"
J10	1KB	"	8	9	"	"	"	"
J11	2KA	"	13	15	"	"	"	"
J12	2KB	"	18	19	"	"	"	"
J13	4KA	"	38	36	"	"	"	"
J14	4KB	"	33	32	"	"	"	"
J15	3KA	"	28	26	"	"	"	"
J16	3KB	"	23	22	"	"	"	"

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NOTES:
1. R1 = 20Ω ±5%, 5 W.
2. S1 = SPDT switch.
3. Item V04-2 must be connected to minimize voltage drop in current carrying leads to item P07-2.
4. Place switch S1 in position A for test conditions J1-J8 and in position B for test conditions J9-J16.
5. Adjust item P07-2 to give a reading of 10.0 ±0.1 V (0.5 A) with point A shorted to point B.
6. Since items V04-1 and V04-2 are not used simultaneously, only one item V04 is required.

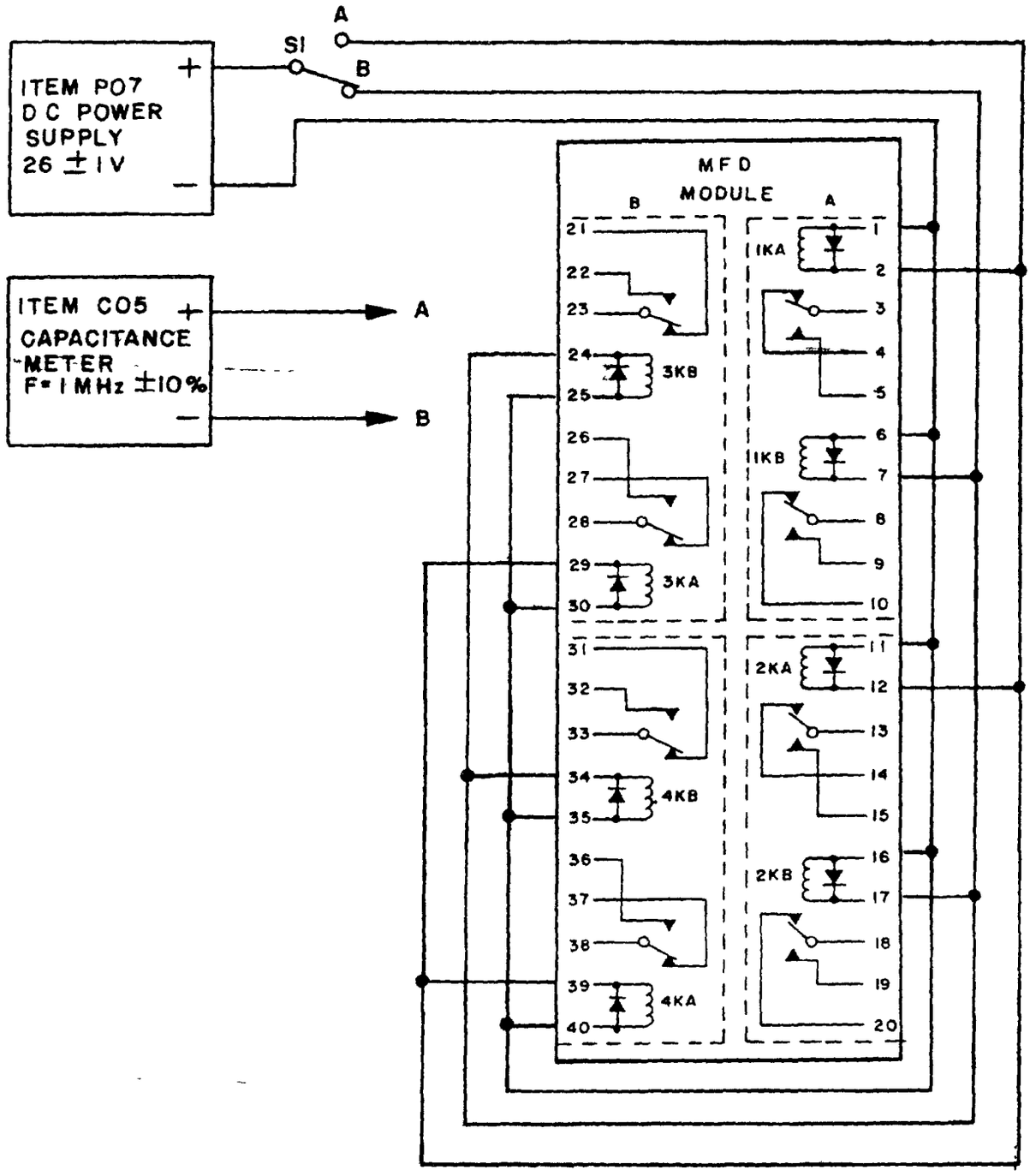
FIGURE 12. Contact carry current check test circuit for table X.

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TABLE XI. Intercontact capacitance test.

Test condition	Parameter		Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
K1	1KA	Intercontact capacitance	3	5	0	5	pF	13
K2	1KB	"	8	9	"	"	"	"
K3	2KA	"	13	15	"	"	"	"
K4	2KB	"	18	19	"	"	"	"
K5	4KA	"	38	36	"	"	"	"
K6	4KB	"	33	32	"	"	"	"
K7	3KA	"	28	26	"	"	"	"
K8	3KB	"	23	22	"	"	"	"
K9	1KA	"	3	4	"	"	"	"
K10	1KB	"	8	10	"	"	"	"
K11	2KA	"	13	14	"	"	"	"
K12	2KB	"	18	20	"	"	"	"
K13	4KA	"	38	37	"	"	"	"
K14	4KB	"	33	31	"	"	"	"
K15	3KA	"	28	27	"	"	"	"
K16	3KB	"	23	21	"	"	"	"

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NOTES.
1. S1 = SPDT switch.
2. Place switch S1 in position B for test conditions K1-K8 and in position A for test conditions K9-K16.
3. Leads A and B to be compensated for stray capacitance and to be as short as possible.

FIGURE 13. Intercontact capacitance test circuit for table XI.

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